



Published in final edited form as:

Neurorehabil Neural Repair. 2025 January ; 39(1): 74–86. doi:10.1177/15459683241287731.

IpsiHand Brain-Computer Interface Therapy Induces Broad Upper Extremity Motor Rehabilitation in Chronic Stroke

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Abstract

Background: Chronic hemiparetic stroke patients have very limited benefits from current therapies. Brain-computer interface (BCI) engaging the unaffected hemisphere has emerged as a promising novel therapeutic approach for chronic stroke rehabilitation.

Objectives: This study investigated the effectiveness of contralesionally-controlled BCI therapy in chronic stroke patients with impaired upper extremity motor function. We further explored neurophysiological features of motor recovery driven by BCI. We hypothesized that BCI therapy would induce a broad motor recovery in the upper extremity, and there would be corresponding changes in baseline theta and gamma oscillations, which have been shown to be associated with motor recovery.

Methods: Twenty-six prospectively enrolled chronic hemiparetic stroke patients performed a therapeutic BCI task for 12 weeks. Motor function assessment data and resting state

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Trial Registration: Advarra Study:

<https://classic.clinicaltrials.gov/ct2/show/NCT04338971>

Washington University Study:

<https://classic.clinicaltrials.gov/ct2/show/NCT03611855>

Disclosures

E.C.L. owns stock in Neuroolutions, Inner Cosmos and Sora Neuroscience. Washington University owns stock in Neuroolutions. This work and E.C.L. have had their conflict of interest rigorously evaluated and managed throughout this study and with the creation of this manuscript. Other authors have nothing to disclose.

electroencephalogram (EEG) signals were acquired before initiating BCI therapy and across BCI therapy sessions. The Upper Extremity Fugl-Meyer assessment (UEFM) served as a primary motor outcome assessment tool. Theta-gamma cross-frequency coupling (CFC) was computed and correlated with motor recovery.

Results: Chronic stroke patients achieved significant motor improvement in both proximal and distal upper extremity with BCI therapy. Motor function improvement was independent of Botox application. Theta-gamma CFC enhanced bilaterally over the C3/ C4 motor electrodes and positively correlated with motor recovery across BCI therapy sessions.

Conclusions: BCI therapy resulted in significant motor function improvement across the proximal and distal upper extremities of patients, which significantly correlated with theta-gamma CFC increases in the motor regions. This may represent rhythm-specific cortical oscillatory mechanism for BCI-driven rehabilitation in chronic stroke patients.

Keywords

Brain-Computer Interface; Chronic Stroke Rehabilitation; Theta-Gamma Cross-Frequency Coupling

Introduction

Until recently, approaches for chronic stroke rehabilitation provided limited therapeutic benefits. Historically, motor improvements were thought to plateau at three months following stroke, leading to permanent motor deficits¹⁻³. However, recent studies employing Brain-Computer Interface (BCI) strategies for motor rehabilitation have shown that significant functional improvements can be attained even during the chronic phase of stroke⁴⁻⁷. Recently, the United States Food and Drug Administration (FDA) authorized the Neuroolutions IpsiHand Upper Extremity Rehabilitation System (IpsiHand System), which is the first BCI therapy for chronic stroke-induced motor impairment. The IpsiHand System uses electroencephalographic (EEG) signals from the unaffected side of the brain to control a distal wearable exoskeleton around the affected hand. Early evidence has shown that regular use of the contralesionally-controlled BCI can facilitate motor recovery in chronic stroke patients⁸⁻¹⁰.

The involvement of the unaffected hemisphere in stroke recovery remains uncertain. Decreases in contralesional activity were associated with optimal recovery in some studies, which suggests that the contralesional hemisphere impedes recovery¹¹⁻¹⁴. Other studies have shown that increased activity in the contralesional hemisphere is associated with improved motor function^{15, 16}. Given the inverse correlation between motor recovery and the extent of corticospinal tract transection¹⁷, we hypothesized that utilizing contralesional hemisphere activity to drive a BCI-controlled exoskeleton could potentially result in functional enhancements. The utilization of the uninjured motor cortex as the signal substrate for BCI motor rehabilitation has further reinforced the positive contribution of the unaffected hemisphere in motor recovery⁹. A BCI provides very precise temporal coupling of the cortical physiology associated with motor intention from the uninjured hemisphere to the sensory feedback provided to the hand. This BCI-enhanced cortico-sensory coupling

is thought to create a Hebbian state to facilitate neuroplastic changes that can improve the brain's control of the impaired hand. The cortical signals used by the IpsiHand System are low-frequency thalamocortical rhythms in the mu (8-12 Hz) and beta (13-29 Hz) bands that are thought to enable thalamic modulation of the cortex^{18, 19}. Given that these rhythms have a broad cortical distribution²⁰, it remains unclear to what extent the neuroplastic changes induced by the IpsiHand are specific to the hand or whether they have a wider functional effect. Given the emerging nature of this intervention, more data is necessary to define better the magnitude and extent of the benefit provided to these patients.

The coupling between various frequency bands could serve as a potential mechanism for motor learning. Recently, there has been growing interest in exploring complex neural oscillation properties, such as cross-frequency coupling (CFC), which involves synchronization between low-frequency oscillation phase and higher-frequency oscillation amplitude²¹⁻²⁴. The mechanisms of learning have been extensively investigated in the hippocampus, with theta-gamma CFC proposed as a pivotal mechanism related to learning²⁵⁻²⁸. It has been shown that CFC similarly plays a role in learning across neocortical regions^{24, 29}. Enhancement of theta-gamma coupling through transcranial alternating current stimulation (tACS) over motor cortex during motor learning has shown significant improvements in motor skill acquisition³⁰. Moreover, theta-gamma coupling has been demonstrated to correlate with functional improvements in chronic stroke rehabilitation³¹. These findings suggest a potential involvement of theta-gamma CFC in motor skill learning.

Here, we aimed to explore the effectiveness of the IpsiHand System for chronic stroke patients with impaired upper extremity motor function. Given that the IpsiHand System is controlled by thalamocortical motor rhythms, we hypothesized that the entire arm would demonstrate a functional improvement. We specifically aimed to assess the global improvement in upper extremity function and also to determine whether the functional benefit was regionally specific (distal limb only) or more general (distal and proximal limb). Further, we assessed the correlation between functional improvements and theta-gamma coupling, a potential mechanism previously implicated in motor recovery. In aggregate, this study demonstrates the robust benefit that the IpsiHand System provides to the entire upper limb and the associated cortical dynamics.

Methods

Study Participants

This study adopted a prospective design, wherein the registration date documented in the trial registry preceded the enrollment date as recorded in the trial registry. One hundred chronic stroke patients with upper-limb hemiparesis consented to participate in the study. Thirty patients completed BCI therapy for 12 weeks (see Supplementary Figure 1). The high number of dropouts in our study can be attributed to the trial requirements and the influence of the COVID-19 pandemic. Thirty-one patients did not meet inclusion/exclusion criteria. Fifteen patients were unable to consistently generate identifiable BCI control signals across two EEG screenings. The trial was heavily impacted by COVID-19, leading to the withdrawal of 24 patients. Inclusion criteria were stroke at least 6 months prior to this study; unilateral upper extremity weakness. We enrolled patients with varying degrees of

hemiparesis, from mild to severe, to investigate the impact of BCI therapy on motor recovery based on the severity of baseline motor impairment. Exclusion criteria were severe visual, cognitive or psychiatric impairment; severe aphasia, ataxia, unilateral neglect; impaired tactile or proprioceptive sensation in the affected upper extremity. We additionally excluded four patients due to their enrollment before the official start date of the study. Twenty-six prospectively enrolled patients from two studies with the same experimental protocol were included in the manuscript (Advarra Study: [ClinicalTrials.gov](https://clinicaltrials.gov/ct2/show/study/NCT04338971) Identifier [NCT04338971](https://clinicaltrials.gov/ct2/show/study/NCT04338971), approval number 201506124, 16 patients; Washington University Study: [ClinicalTrials.gov](https://clinicaltrials.gov/ct2/show/study/NCT03611855) Identifier [NCT03611855](https://clinicaltrials.gov/ct2/show/study/NCT03611855), approval number 201701021, 10 patients). The Advarra and Washington University cohorts comprised chronic stroke patients, who were prospectively enrolled between May 5, 2017 - October 9, 2020, and April 26, 2018 - March 18, 2020, respectively. Motor function outcomes were primarily evaluated using the validated Upper Extremity Fugl-Meyer assessment (UEFM). We also assessed UEFM section scores to measure motor function improvements across the proximal and distal upper extremities (refer to Supplementary Table 1 for the methodology used to calculate UEFM section scores)³². Secondary measures included the Action Research Arm Test (ARAT), Arm Motor Ability Test (AMAT), Motricity Index, Gross Grasp (Hand Grip Dynamometer), Modified Ashworth Scale (MAS) at the wrist and elbow. Patients received Botox injections, which coincided with their BCI therapy (Supplementary Table 2). Thus, we further investigated if motor function improvement following BCI therapy was significantly influenced by Botox application. The written informed consent process followed Good Clinical Practice Guidelines. All questions were addressed, and patients verbalized understanding prior to signing informed consent.

BCI System Design

The IpsiHand System consisted of a wireless EEG headset with six active electrodes, EEG amplifier, and robotic hand orthosis (the FDA-approved IpsiHand Upper Extremity Rehabilitation System, Neuroolutions, Santa Cruz, CA, USA) (Figure 1A). Bluetooth-enabled touchscreen tablet was connected to the EEG headset to record brain signals. Local Wi-Fi network facilitated communication between the tablet and orthosis. The tablet guided patients through BCI tasks and converted changes in spectral power into control signals for the orthosis to open and close it in a 3-finger pinch grip. During the BCI task, patients were instructed to open the orthosis through motor imagery of their affected hand or to maintain it closed by resting quietly. The orthosis opened and closed in response to alterations in the power of the patient-specific control signal. Subjects who could partially move their affected arm were instructed to allow passive movements facilitated by the orthotic device.

Intervention Protocol

Figure 1B illustrates the BCI therapy timeline. Prior to the therapeutic intervention, a screening session was conducted where patients performed a series of rest and motor imagery trials. The BCI device control signal was determined based on 1 Hz width frequency band with spectral power modulation that best aligned with the differences between rest and motor trials. The control signal was within the mu (8-12 Hz) or beta (13-29 Hz) frequency band and remained unchanged for each patient throughout BCI therapy. Only patients who exhibited a consistent feature frequency were included in the study. Baseline

motor function was assessed by physical and occupational therapists before commencing the therapy. Research team members then provided training to patients on how to use the BCI system. Patients were instructed to use the device for one hour/day, five days/week, over the course of 12 weeks. To avoid the observer bias, motor assessment was performed by the same clinician per patient before commencing BCI therapy and after completion of a BCI therapy session at four-week intervals. Patients did not receive concomitant occupational or physical therapy while undergoing the BCI therapy over the 12-week period.

Each session of BCI therapy had a duration of approximately one hour and comprised one calibration phase followed by five BCI therapy runs. The pre-therapy calibration was employed to ensure data quality and identify motor imagery activity during the BCI task. During calibration, patients rested quietly and then proceeded to complete a series of task blocks and rest trials. Within the task blocks, patients were instructed to mentally envision moving their affected hand. The orthosis remained stationary during calibration. After the calibration phase, patients initiated the BCI therapy runs. Each run consisted of 30 motor imagery and 30 rest trials, with the trial order randomized. A three-second "fidget" period was included between each eight-second trial, encouraging patients to blink or make minor physical adjustments. Upon completing a BCI therapy run, the system paused, allowing patients to rest before resuming their therapy. Resting state EEG data from pre-task calibration sessions were stored for subsequent analysis.

Electroencephalogram Recording and Processing

EEG data were acquired using six wireless dry electrodes (F3, F4, C3, C4, P3, and P4) mounted on an EEG headset in the International 10–20 System (Neuroolutions, Santa Cruz, CA, USA). The EEG signals were sampled at 300 Hz, and ground electrode was placed on the forehead. Electrode impedance was maintained below 10 k Ω . EEG data obtained during the pre-therapy calibration rest period were analyzed across BCI therapy sessions (Pre-BCI, Post-BCI 4th, 8th and 12th week; each five minutes long). The EEG recordings underwent band-pass filtering between 1 to 100 Hz. A 60 Hz notch filter was applied to eliminate environmental noise. The independent component analysis (ICA) was applied for artifact attenuation³³⁻³⁵. Independent components associated with eye blinks, lateral eye movements, muscle-related activity, and cardiac artifacts were identified and eliminated. The EEG data were re-referenced using a common average reference.

Prior research has shown that non-invasive scalp EEG can detect alterations in high gamma power associated with motor tasks³⁶. Scalp EEG was found to effectively record high gamma activity up to 100 Hz^{22, 37-40}. We conducted time-frequency analysis to visualize resting high gamma cortical oscillations³⁹. EEG was filtered offline using a FIR band pass filter from 65 to 100 Hz. Data were segmented into 5 sec epochs. A Morlet wavelet convolution was computed using the channel time–frequency option. Thirty-five linearly spaced frequencies were computed between 65 and 100 Hz. For each patient, time–frequency data were averaged across all epochs per condition. The grand average time–frequency maps were obtained by averaging data across patients (see Supplementary Figure 2A, **top panels**). High gamma oscillations were then averaged across 65-100 Hz to visualize a single high gamma frequency wave (see Supplementary Figure 2A, **bottom panels**).

Cross-Frequency Coupling

CFC was calculated on resting EEG data, during which patients rested quietly and the orthosis was not active. The raw EEG recording was band-pass filtered in the theta and gamma frequency bands. The data were segmented into five sec epochs. We applied a Hilbert transform to derive the complex-valued analytic signal. Estimates of theta band phase and gamma band amplitude were extracted from the analytic signal. The coupling between theta band (4-8 Hz) phase and gamma band (65-100 Hz) amplitude was calculated using the Mean Vector Length (MVL) approach²³. For each patient, CFC values were averaged across all epochs. The grand average CFC plot was obtained by averaging data across patients. Theta-gamma CFC was computed across BCI therapy sessions and compared between sessions.

As a complimentary tool, Canolty maps were computed to illustrate the theta-gamma coupling²³. The phase troughs of the low frequency were specified at theta frequency band (5 Hz). A time window of 1 sec was extracted around each of these troughs. A time-frequency decomposition of these short epochs was performed. The power of all the time-frequency maps was averaged to obtain the final Canolty maps (see Supplementary Figure 2B). This approach enabled us to visualize whether high-frequency power systematically varied systematically with the phase of low frequencies, known as CFC.

Statistical Analyses

Changes in motor assessment scores with BCI therapy were assessed in a repeated-measures Analysis of Variance (ANOVA) with within-subjects factor *Session* (Pre-BCI, Post-BCI 4th, 8th and 12th week). We hypothesized that BCI therapy would induce motor recovery in the upper extremity. Planned contrasts were used to decompose differences in motor scores between Pre-BCI versus Post-BCI sessions. Subsequent contrasts were employed to examine differences in motor scores between Post-BCI sessions. A nonparametric permutation test was performed to examine the differences in mean CFC values following BCI therapy⁴¹. Nonparametric Spearman rank correlation analyses were conducted between motor assessment scores and theta-gamma CFC values. Correlations were also computed between motor assessment scores and BCI usage/performance data. In all statistical tests, the significance thresholds were set at 0.05, and the P-values underwent adjustment using a Bonferroni correction.

Data Availability

The data will be made available upon reasonable request to the corresponding author.

Results

BCI usage and performance

BCI usage data and the success rates of BCI trials are detailed in Supplementary Table 3. Patients had to enter their usage on a provided tracking sheet, which assisted them in documenting therapy times. Therapy dosage for patients was estimated by summing the number of runs with at least 10% accuracy on both movement imagery and rest trials. The

usage of BCI varied among patients, with an average of 41.6 ± 18.7 hours (mean $\pm$ SD). There were no adverse events.

Patients generally used their BCI systems effectively, achieving median move and rest success rates of 71.8% and 39.8%, respectively. A definition of a successful trial was reaching the BCI activation threshold for at least 1 second for move trials or staying under the activation threshold for the entire trial duration for rest trials. Most patients achieved higher success rates with movement imagery trials due to the more stringent criteria for success in rest trials.

Patient characteristics

Patient characteristics and primary motor assessment scores are shown in Table 1. The age was 58.4 ± 11.8 years (mean $\pm$ SD), 7 females and 19 males. Overall, 14 patients had left hemispheric stroke, and 12 patients had right hemispheric stroke. This study included patients with an average post-stroke duration of 69.1 ± 82.9 (mean $\pm$ SD) months.

Motor Assessment Scores

The UEFM served as a primary motor outcome assessment tool. Patients achieved an increase in UEFM score following BCI therapy. The mean increase in UEFM score was 8.1 points which exceeded the minimal clinically significant difference (MCID) threshold of 5.25 points increase⁴². A total of 18 out of the 26 patients reached the MCID. None of the patients exhibited a decline in motor function. Both the Advarra and Washington University studies demonstrated significant motor improvements with BCI intervention, and there were no significant differences in terms of motor recovery between the two studies (see Supplementary Figure 3). The final changes in UEFM scores for the Advarra and Washington University Studies were 8.8 and 7.1 points, respectively, with a p-value of 0.38.

Primary and secondary motor assessment scores across BCI therapy sessions were presented in Figures 2 and 3. **UEFM:** the main effect of session was significant, $F(3,75) = 32.66$, $p < 0.001$. Planned contrasts revealed significant increases in UEFM scores. **ARAT:** the main effect of session was significant, $F(3,75) = 16.32$, $p < 0.001$. Planned contrasts revealed significant increases in ARAT scores. **AMAT:** the main effect of session was significant, $F(3,75) = 26.13$, $p < 0.001$. Planned contrasts revealed significant increases in AMAT scores. **Motricity Index:** the main effect of session was significant, $F(3,75) = 19.48$, $p < 0.001$. Planned contrasts revealed significant increases in motricity index scores. **Gross Grasp:** the main effect of session was significant, $F(3,75) = 5.06$, $p = 0.003$. Planned contrasts revealed significant increases in motricity index scores. **Elbow and Wrist MAS:** the main effect of session did not prove significant, $F(3,75) = 2.06$ and 0.02 , $p = 0.11$ and 0.92 , respectively, excluding significant changes across BCI therapy sessions.

UEFM score changes showed significant positive correlations with BCI usage and move trial success rates (see Figure 4). There were no significant correlations between final UEFM score changes after 12 weeks of BCI treatment and baseline UEFM scores ($r = -0.22$, $p = 0.28$), which excludes the differential effects of BCI therapy on motor recovery depending on the severity of baseline motor impairment.

Upper Extremity Fugl-Meyer Section Scores

We evaluated changes in UEFM section scores following BCI therapy (see Supplementary Table 1 for UEFM section description). The results are summarized in Figure 5A. **Section A1** (triceps/biceps reflex): the main effect of session was not significant, $F(3,75) = 2.45$, $p = 0.07$, excluding significant changes across BCI therapy sessions. **Sections A2, A3 and A4** (proximal upper extremity): the main effect of session was significant, $F(3,75) = 15.99$, 3.42 and 6.15 , $p < 0.001$, 0.02 and 0.002 , respectively. Planned contrasts revealed significant increases in UEFM section scores. **Sections B and C** (distal upper extremity): the main effect of session was significant, $F(3,75) = 9.17$ and 10.29 , $p < 0.001$ and 0.001 , respectively. Planned contrasts revealed significant increases in UEFM section scores. **Section D** (whole limb coordination): the main effect of session was significant, $F(3,87) = 5.25$, $p = 0.003$. Planned contrasts revealed significant increases in UEFM section scores.

Comparison of motor recovery between Botox and Non-Botox groups

Both Botox and Non-Botox patients achieved greater than MCID gains in UEFM scores (10.7 and 6.9 points gain, respectively). The proportion of patients in each group who met the MCID for UEFM was similar: 67% (6/9) of Botox patients and 71% (12/17) of Non-Botox patients. Further statistical analyses were performed to assess if the influence of Botox application on motor recovery following BCI therapy was significant. Pre-BCI baseline motor scores and final motor score change (motor score Post-BCI 12th week - Pre-BCI baseline motor score) were compared between Botox ($n = 9$) and Non-Botox groups ($n = 17$) (Figures 3B and 5B). The differences in baseline motor scores and final motor score change between groups were not statistically significant (all p 's > 0.05), except baseline Elbow MAS score. However, although baseline Elbow MAS score was significantly lower in Botox group compared with Non-Botox group ($p = 0.03$), groups did not significantly differ in final Elbow MAS score change ($p = 0.51$) (Figure 3B). The Botox and Non-Botox groups were also compared for the UEFM section scores. The results presented in Figure 5B indicate that neither baseline motor scores nor final motor score changes were statistically different between groups for any section of the UEFM (all p 's > 0.05).

Modulation of brain activity

Theta-gamma cross-frequency coupling—Supplementary Figure 4A depicts topographic representation of mean theta-gamma CFC values across BCI therapy sessions (Pre-BCI, Post-BCI 4th, 8th and 12 week). We compared theta-gamma CFC values between Pre-BCI baseline and Post-BCI sessions as measured over all the electrode sites (Supplementary Figure 4B). During the earlier post-treatment session (Post-BCI 4th week), theta-gamma CFC increased at the C3 and C4 motor electrodes. This was in the expected direction but did not reach significance (both p 's > 0.05). The subsequent Post-BCI sessions were associated with significant bilateral enhancement of theta-gamma CFC over the motor electrodes (Post-BCI 8th week: significant effects at the C3 and C4 electrodes, both p 's < 0.05 ; Post-BCI 12th week: significant effects at the C3 and C4 electrodes, both p 's < 0.05). Post-BCI 12th week vs. Post-BCI 8th week contrast did not reveal significant difference in theta-gamma CFC (all p 's > 0.05). Theta-gamma CFC over the frontal and parietal electrodes was not significantly modulated throughout BCI therapy (all p 's > 0.05).

Correlations between motor recovery and theta-gamma cross-frequency coupling

Correlations between motor recovery and theta-gamma CFC change at the C3 and C4 electrodes across BCI therapy sessions relative to baseline are shown in Supplementary Figure 5. UEFM, AMAT and Motricity Index score changes showed significant positive correlations with theta-gamma CFC change. ARAT, Gross Grasp, Elbow and Wrist MAS score changes did not correlate significantly with theta-gamma CFC change. We also conducted correlation analyses for UEFM section scores (Supplementary Figure 6). The gains in both proximal and distal UEFM section scores correlated significantly with theta-gamma CFC change at the C3 and C4 electrodes.

Discussion

This study provides compelling evidence demonstrating that the use of a contralesionally-controlled BCI for chronic stroke therapy, the IpsiHand System, induced clinically significant motor function improvement. Despite being more than six months post-stroke, patients with chronic stroke achieved a significant increase in UEFM score. Importantly, we found improvement in both proximal and distal upper extremity section scores of UEFM. Additionally, improvements were observed in secondary measures of motor function. A significant increase was evident in ARAT, AMAT, Motricity Index and Gross Grasp scores, which collectively assess proximal and distal upper extremity motor function. Conversely, the measurements of muscle spasticity, such as Elbow and Wrist MAS, did not exhibit significant changes following BCI intervention. These findings further corroborate previous evidence, reinforcing the notion that BCI-controlled rehabilitation systems can facilitate motor recovery^{5, 7}. However, this study was unique in utilizing the unaffected hemisphere in a BCI rehabilitation system and demonstrating the benefit for improvement is widespread in the affected limb, setting it apart from previous research. Our findings provide support for the engagement of the unaffected hemisphere as an important substrate for motor rehabilitation.

According to a meta-analysis, five out of nine BCI groups and three out of nine control groups, which also included conventional therapy, showed motor improvements exceeding the minimal clinically important difference (MCID > 5.25)¹⁰. However, the studies that reported clinically meaningful motor function gains had notably smaller sample size, ranging from as low as 7 to as high as 15 patients, in contrast to the 26 patients in our study. Furthermore, the time since stroke in those studies was considerably shorter, ranging from 2.2 ± 1.7 (mean $\pm$ SD) to 37.3 ± 43.9 (mean $\pm$ SD) months, whereas our study included patients with an average post-stroke duration of 69.1 ± 82.9 (mean $\pm$ SD) months. There is also heterogeneity among studies stemming from variations in patient characteristics, such as age, gender, and treatment adherence and duration, which complicates direct comparisons.

In this study, both Botox and Non-Botox groups achieved greater than MCID gains in UEFM scores. In addition, the proportion of patients in each group who met the MCID for UEFM was nearly the same. BCI therapy provided improvement in both proximal and distal upper extremity section scores of UEFM. The gains for each specific UEFM section were also not significantly different between Botox and Non-Botox groups. These results

indicate that Botox application did not have a statistically significant impact on the benefit provided by the BCI therapy. The use of Botox to reduce spasticity may have functional effects that both facilitate and impair motor recovery. In a previous study, Botox application for post-stroke upper limb spasticity did not improve upper limb motor function, including activities such as grasping and reaching⁴³. Botox has also been associated with increased motor impairment, including weakness of injected muscles⁴⁴. Thus, while reducing muscle tone can enhance finer movements, it is important to acknowledge that it may also decrease strength and hinder overall motor function. Regardless, the presence of Botox did not facilitate or hinder the effect of the IpsiHand System's functional benefits.

In addition to the functional gains associated with the use of the IpsiHand System, we also evaluated whether there were concomitant cortical electrophysiologic changes associated with these motor improvements. Given prior work that showed theta-gamma CFC was associated with BCI-driven motor rehabilitation³¹, we further evaluated the dynamic changes in theta-gamma CFC throughout BCI therapy to evaluate the neurophysiological mechanisms underlying motor recovery. A key electrophysiologic finding of our study was the bilateral enhancement of coupling between theta and gamma oscillations over the motor areas following BCI therapy. This effect showed a significant positive correlation with both proximal and distal upper limb motor improvement. While the exact functional significance of CFC remains uncertain, it is believed to play a crucial role in coordinating neuronal activity across brain regions^{21, 45}. During the learning of item-context associations, theta-gamma coupling showed positive correlation with behavioral performance accuracy suggesting its involvement in learning process²⁵. Theta-gamma coupling has been shown to emerge in the primary motor cortex when gamma-aminobutyric acidergic (GABAergic) activity is inhibited⁴⁶. Reduced GABAergic activity in the primary motor cortex represents a central mechanism for motor plasticity⁴⁷. This study suggests that enhanced synchronization of theta and gamma oscillations may play a mechanistic role in motor recovery.

Limitations

This study was conducted with the assumption that motor deficits would remain stable in the chronic stage of stroke. Indeed, previous research has shown limited improvement in motor deficits during the chronic stage of stroke^{2, 3, 48, 49}. Thus, we did not have a control group for comparison with BCI therapy. Furthermore, in a separate study investigating motor rehabilitation in stroke patients, sham BCI therapy failed to facilitate motor recovery⁵. Consequently, we primarily attribute the observed improvement in motor function and associated electrophysiological findings in this study to the BCI intervention. However, we cannot currently determine whether enhanced theta-gamma coupling is specific to BCI techniques, or it represents a broader phenomenon associated with other methods utilized for chronic stroke rehabilitation. Additionally, because assessors were not blinded, there was a potential risk of assessor bias when evaluating the changes in motor outcomes over time.

Conclusions

IpsiHand BCI therapy resulted in significant improvement in both proximal and distal upper extremity motor function of chronic stroke patients, which was independent of Botox application. Motor recovery was paralleled with enhanced theta-gamma coupling over the motor regions, which may represent rhythm-specific cortical oscillatory mechanisms for BCI-driven motor rehabilitation.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

Funding

This work was supported by National Institutes of Health (NIH) R21NS102696 (E.C.L. and A.C.) and National Institute of Biomedical Imaging and Bioengineering (NIBIB) P41-EB018783 (E.C.L.).

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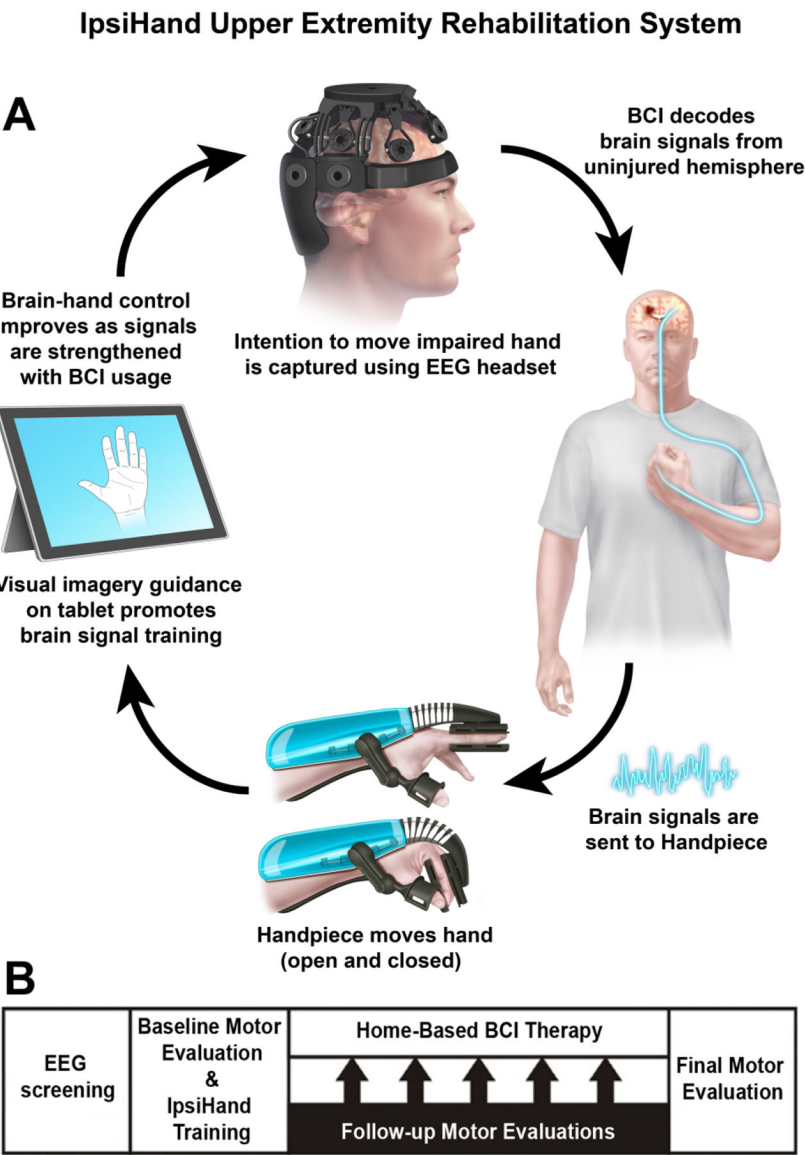


Figure 1. Experimental design. (A) BCI System Design. Patients performed motor imagery tasks. Contralateral EEG signals were decoded and translated into commands to open or close the orthosis. The orthosis provided proprioceptive sensory feedback. (B) 12-week BCI Therapy Protocol. Eligible patients underwent motor function assessments and resting EEG recordings prior to commencing BCI therapy and at four-week intervals after completion of a BCI therapy session. BCI: Brain-Computer Interface.

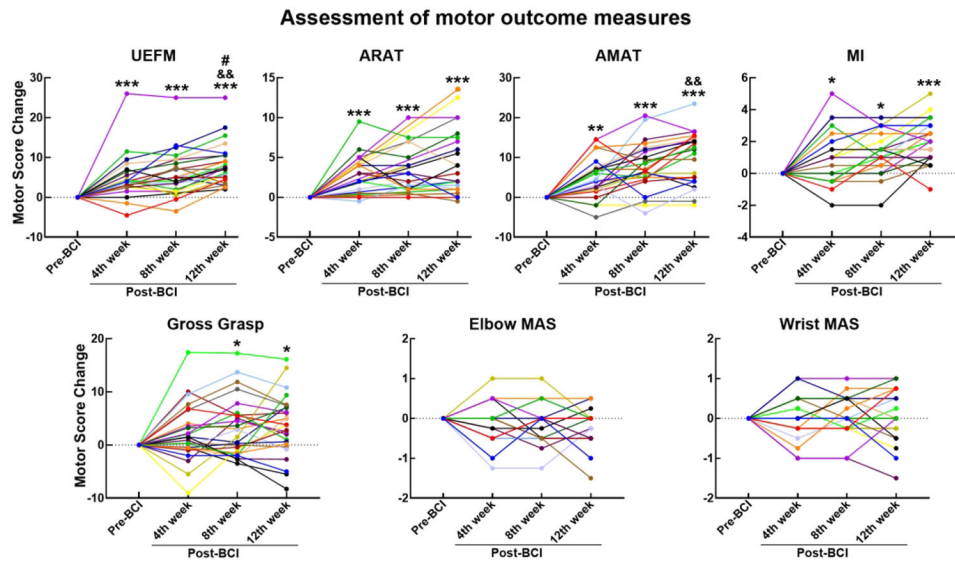


Figure 2. Motor assessment score changes.

Y-axis, motor score change; X-axis, BCI therapy sessions. UEFM: Upper Extremity Fugl-Meyer; ARAT: Action Research Arm Test; AMAT: Arm Motor Ability Test; MAS: Modified Ashworth Scale. Patients are depicted in different colors. Significance levels were based on the *Pairwise Comparisons in ANOVA* (Bonferroni corrected). *, **, *** symbols: $p \leq 0.05$, 0.01 and 0.001 for Pre-BCI vs. Post-BCI 4th, 8th and 12th week contrasts; && symbol: $p \leq 0.01$ for Post-BCI 4th vs. 12th week contrasts; # symbol: $p \leq 0.05$ for and Post-BCI 8th vs. 12th week contrast; BCI: Brain-Computer Interface.

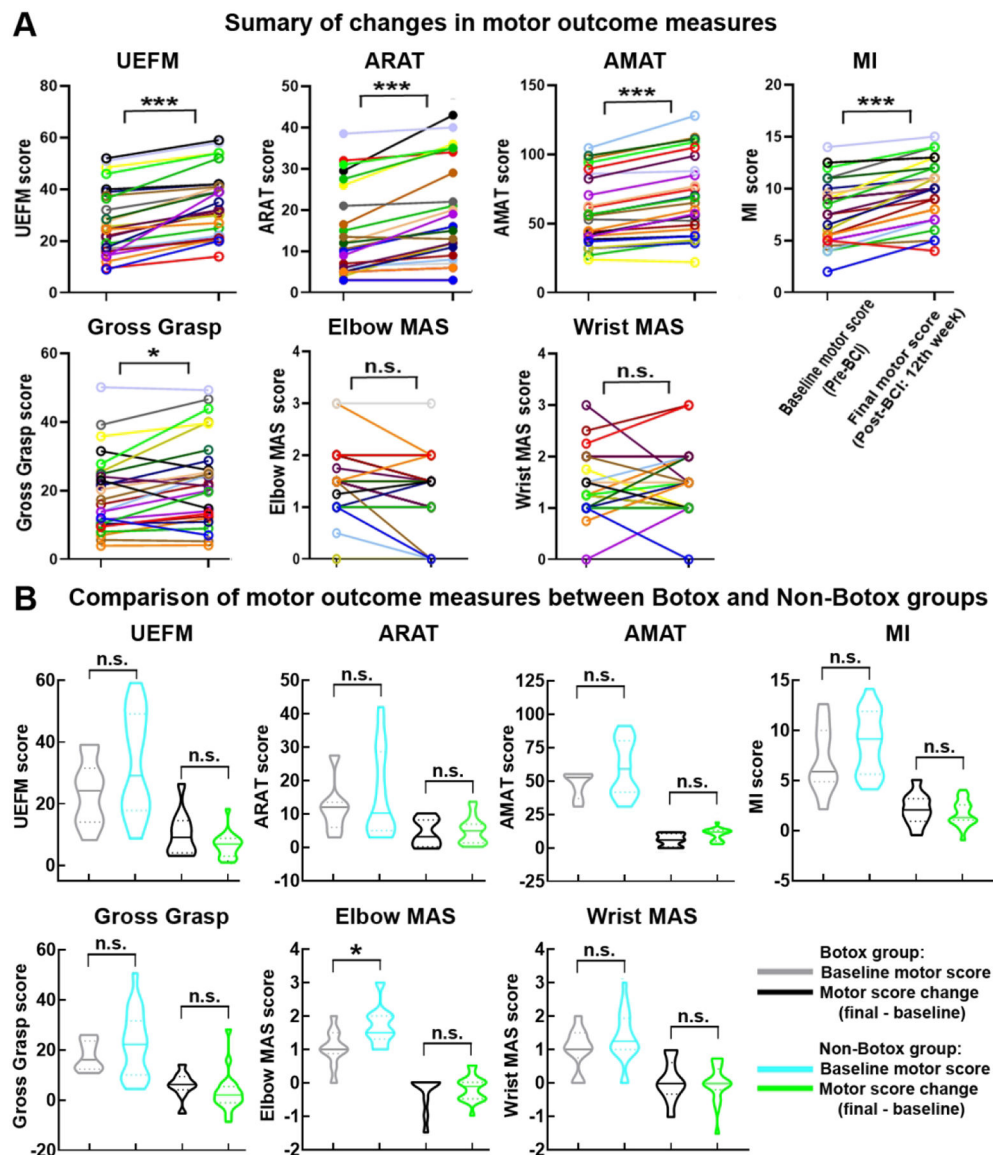


Figure 3. Summary of motor recovery measures.

(A) Baseline and final motor scores. Patients are depicted in different colors. **(B)** Botox and Non-Botox groups. Baseline motor scores and final motor score changes. Y-axis, motor score; X-axis, BCI therapy sessions. UEFM: Upper Extremity Fugl-Meyer; ARAT: Action Research Arm Test; AMAT: Arm Motor Ability Test; MAS: Modified Ashworth Scale; BCI: Brain-Computer Interface. *, *** symbols: $p \leq 0.05$, and 0.001 ; n.s.: not statistically significant.

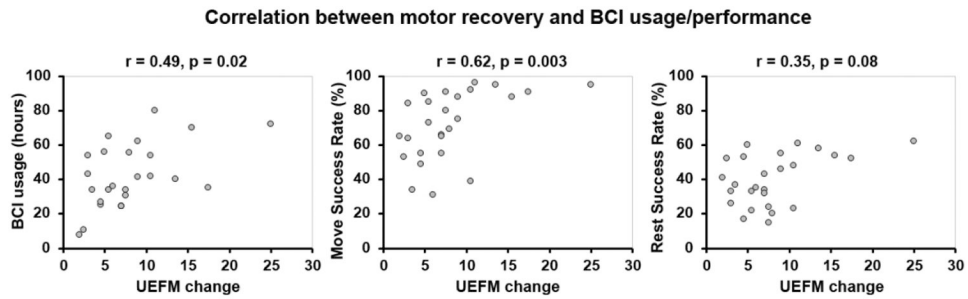


Figure 4: Correlation between UEFM scores and BCI usage/performance.

Spearman rank correlations were run between changes in UEFM scores after 12 weeks of BCI treatment and BCI Usage, Move Success Rate, Rest Success Rate. Y-axis, UEFM score change; X-axis, BCI usage or performance. UEFM: Upper Extremity Fugl-Meyer; BCI: Brain-Computer Interface.

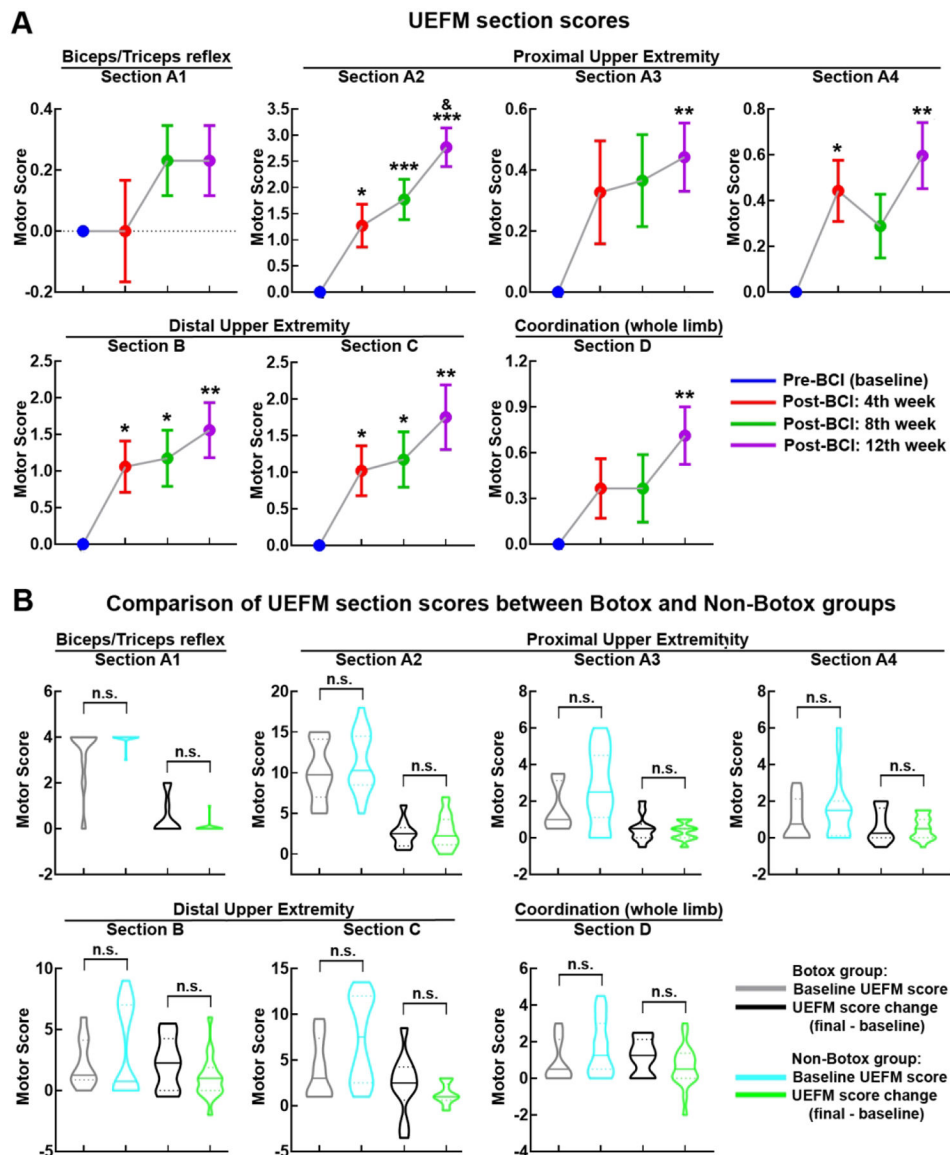


Figure 5. Section scores of UEFM.

(A) UEFM section scores across BCI therapy sessions. **(B)** Botox and Non-Botox groups. Baseline UEFM section scores and final UEFM section score changes. Y-axis, motor score; X-axis, BCI therapy sessions. UEFM: Upper Extremity Fugl-Meyer; BCI: Brain-Computer Interface; *, **, *** symbols: $p \leq 0.05$, 0.01 and 0.001 for Pre-BCI vs. Post-BCI 4th, 8th and 12th week contrasts; & symbol: $p \leq 0.05$ for Post-BCI 4th vs. 12th week contrast; n.s.: not statistically significant.

Table 1:

Patient characteristics and primary motor assessment scores.

Patient	Affected UE / Hand Dominance	Time Post-Stroke (months)	Botox Usage (Yes / No)	Baseline UEFM	Final UEFM	UEFM change (Final - Baseline)
1	LUE/ Right	44	Yes	9	20	11
2	LUE/ Right	153	No	9.5	14	4.5
3	RUE/ Right	20	Yes	36.5	52	15.5
4	RUE/ Right	9	Yes	14	39	25
5	RUE/ Right	19	No	24.5	27	2.5
6	LUE/ Right	61	No	52	59	7
7	RUE/ Right	372	Yes	37.5	41	3.5
8	RUE/ Right	24	No	17.5	35	17.5
9	RUE/ Right	20	No	24.5	32	7.5
10	LUE/ Right	22	No	16	21	5
11	RUE/ Right	20	Yes	28.5	39	10.5
12	RUE/ Right	34	Yes	27.5	41	13.5
13	LUE/ Right	253	No	46	54	8
14	LUE/ Right	38	Yes	16.5	22	5.5
15	RUE/ Right	65	No	51	58	7
16	LUE/ Right	34	No	32	39	7
17	LUE/ Right	122	No	48.5	54	5.5
18	LUE/ Right	96	Yes	25.5	30	4.5
19	RUE/ Right	16	No	19	25	6
20	RUE/ Right	28	Yes	14.5	22	7.5
21	LUE/ Right	12	No	12	21	9
22	RUE/ Right	103	No	40	42	2
23	LUE/ Left	27	No	17	20	3
24	RUE/ Right	68	No	39	42	3
25	RUE/ Right	108	No	21.5	32	10.5
26	LUE/ Right	28	No	22	31	9

LUE: left upper extremity; RUE: right upper extremity; UE: upper extremity; UEFM: upper extremity Fugl-Meyer.